BY SOMMER

Building the Constitutional Foundation for Human–Digital Civilization

BY SOMMER is an independent open research initiative by Michael Sommer dedicated to designing the constitutional architecture through which humanity and intelligent systems may evolve together.

Its purpose is not to replace human judgement, institutions, or democratic governance, but to strengthen the relationship between them through trust, transparency, semantic understanding, and responsible technological evolution.

The world is entering an era where intelligent systems will increasingly participate in everyday life, public administration, scientific discovery, emergency response, business, healthcare, education, and planetary stewardship. Their capabilities will continue to evolve. The fundamental question is therefore no longer whether artificial intelligence will become more capable, but whether society can ensure that capability evolves alongside accountability, human dignity, and public trust.

BY SOMMER approaches this challenge through constitutional rather than technological thinking. Instead of building isolated applications, it develops foundational principles, reference architectures, and open frameworks that preserve human intent while enabling increasingly capable digital systems to collaborate responsibly.

At the centre of this architecture stands the Digital Twin—a constitutional digital representation designed to assist, mediate, verify, organize, and preserve continuity on behalf of the individual. It does not replace the human. It extends the individual's ability to navigate an increasingly complex digital society while remaining governed by clearly defined boundaries of authority, consent, explainability, accountability, and human oversight.

Supporting this foundation is AURA, the Semantic Mediator, which transforms human intention into structured, verifiable digital understanding while translating complex computational reasoning back into language that people, organizations, and institutions can understand.

Technology should adapt to humanity—not require humanity to adapt to technology.

BY SOMMER is founded upon a simple conviction:

Trust is not a feature.

Trust is infrastructure.

It is earned through transparency.

Protected through constitutional boundaries.

Strengthened through explanation.

Sustained through accountability.

Preserved through continuous evolution rather than continual replacement.

The long-term ambition is to establish an open, collaborative research ecosystem where academia, governments, industry, civil society, and independent researchers contribute to trustworthy digital infrastructure that serves humanity across generations.

Founded by Michael Sommer, BY SOMMER is not a destination but a continuing research programme dedicated to one enduring question:

How can humanity and intelligent systems continue to evolve together without continuously

rebuilding trust?

Every principle, architecture, and contribution developed through BY SOMMER seeks to answer that question—not for today's technology, but for the generations that follow.